

Guorui Sang

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Research focus: Diffusion models, neural operators, and robot dynamics

EDUCATION

University of Illinois Chicago

Aug 2025 - Present

Ph.D. in Computer Science | Expected graduation: 2030 | GPA: 4.00/4.00

Beijing University of Technology

Sep 2022 - Jul 2025

Master's in Computer Science | GPA: 3.94/4.00

Hainan University

Sep 2017 - Jul 2021

Bachelor's in Software Engineering

PUBLICATIONS

ICLR 2026 · ReALM-GEN Workshop

Hamiltonian-Guided Diffusion Fields for Variable-Length Rigid-Arm Trajectory Generation. **Guorui Sang** and Pedram Rooshenas.

IEEE ICASSP 2025

ConSinger: Efficient High-Fidelity Singing Voice Generation with Minimal Steps. Yulin Song*, **Guorui Sang***, Jing Yu, and Chuangbai Xiao. **Equal contribution.*

RESEARCH EXPERIENCE

University of Illinois Chicago

Aug 2025 - Present

Graduate Research Assistant since May 2026 | Advisor: Pedram Rooshenas

Function-Space Transformer · *Manuscript in preparation*

- Proposed Recursive FST for **operator learning** to preserve local detail and capture long-range dependencies with a compact latent representation. Implemented **input-dependent spatial anchors** and recursive attention over a **continuous feature field**.
- Achieved error comparable to FNO on PDEBench Burgers. On ImageNet-1K, improved top-1 accuracy by **1.73 percentage points** over a ViT baseline in a single-seed comparison using models with about 3M parameters.

Hamiltonian-Guided Trajectory Generation · *Manuscript in preparation*

- Proposed Hamiltonian-guided diffusion to generate **physically plausible robotic trajectories** across variable horizons without a simulator at inference. Formulated an energy function from **Hamiltonian residuals** for **guided sampling**, steering diffusion denoising through gradient guidance and importance resampling.
- Reduced mean combined position and momentum RMSE by **36% to 46%** versus unguided diffusion across three unseen torque policies on a 3-DoF MuJoCo arm, using four-candidate resampling.

Beijing University of Technology

Jul 2024 - Dec 2024

ConSinger · Singing voice synthesis

- Co-developed a consistency model for singing voice synthesis and designed a quality scorer to select the denoising starting point.
- Cut inference time by **65%** versus DiffSinger on PopCS with comparable perceptual quality (MOS: 3.88 vs. 3.81).

TECHNICAL SKILLS

Languages and frameworks: Python, C/C++, PyTorch, PyTorch Lightning

Tools: Git, Linux, MuJoCo, Weights & Biases

TEACHING EXPERIENCE

Teaching Assistant | UIC | Led MATLAB labs for 30+ students

Aug 2025 - May 2026